PES UNIVERSITY

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Cryptography Project

Image Cryptography using AES-256 Algorithm

Submitted by

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PROGRAM B.TECH

OBJECTIVE –

The primary objective of this project is to implement and evaluate an image cryptographic system using the AES-256 symmetric encryption algorithm in Python. The project aims to:

- Encrypt each RGB channel of an image using AES-256 in CTR mode.
- Evaluate encryption security through entropy, MSE, and NPCR.
- Visualize the encryption quality using histogram analysis.
- Ensure strong security against unauthorized access and image tampering.

METHODLOGY-

The project follows a structured approach to implement and evaluate image encryption using AES-256 in CTR mode. The steps involved are:

1. Image Acquisition and Preprocessing
   The user selects an RGB image, which is loaded into the program and converted into a NumPy array. The image is then split into its three separate color channels: Red, Green, and Blue.
2. Key and Nonce Generation
   For each color channel, a unique 256-bit encryption key and an 8-byte nonce are randomly generated using Python's Crypto.Random module. This ensures that each channel is encrypted independently with a secure key-nonce pair.
3. Encryption using AES-256 in CTR Mode
   Each channel is converted into a byte stream and padded to match AES block size. The AES-256 encryption is applied in CTR (Counter) mode, where a counter combined with a nonce produces a keystream that is XORed with the plaintext.
4. Decryption Process
   Using the same key and nonce for each channel, the encrypted data is decrypted back to its original form. The decryption process mirrors the encryption steps and restores the image channels accurately.
5. Image Reconstruction
   The decrypted R, G, and B channels are combined to reconstruct the full RGB image. A slight modification is intentionally introduced in the decrypted image to analyse MSE under imperfect conditions.

6. Security Evaluation
   The encrypted channels are evaluated using the entropy, MSE (Mean Squared Error) and NPCR (Number of Pixels Change Rate).
7. Histogram Analysis
   Histograms are generated for both original and encrypted channels. A uniform histogram in the encrypted image indicates good encryption quality and resistance to visual analysis.
8. Output Storage
   Both the encrypted and decrypted images are saved for comparison and further inspection.

THEORY-

Advanced Encryption Standard (AES)

AES is a symmetric block cipher used to encrypt and decrypt data in blocks of 128 bits. It has key sizes of 128, 192, or 256 bits. AES-256 is known for its high level of security due to the longer key length, making brute-force attacks computationally infeasible.

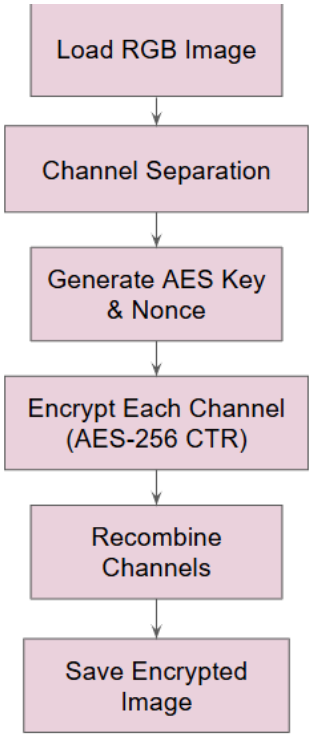
CTR Mode (Counter Mode)

In CTR mode, a counter is combined with a nonce and encrypted with the key to produce a keystream that is XORed with plaintext. It allows AES to behave like a stream cipher, enabling parallel processing and high performance.

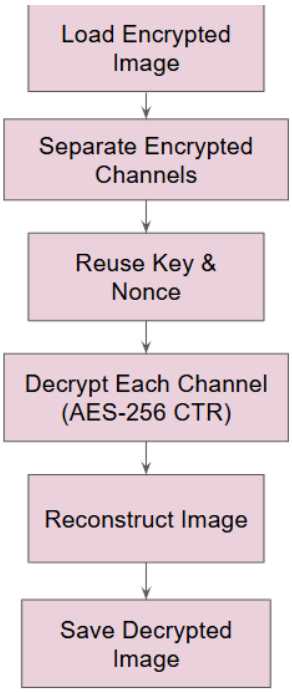
Security Metrics

- **Entropy**: Measures randomness in the encrypted image (ideal value ≈ 8 for 8-bit images).
- **MSE (Mean Squared Error)**: Quantifies the difference between original and decrypted image.
- **NPCR (Number of Pixels Change Rate)**: Measures sensitivity of the cipher to a single pixel change.

Block Diagram:



Encryption process



Decryption Process

Code:

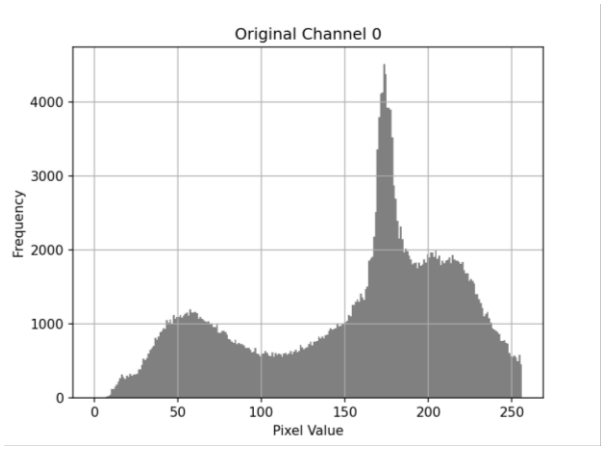
```
import numpy as np
from PIL import Image
from Crypto.Cipher import AES
from Crypto.Util.Padding import pad, unpad
from Crypto.Random import get_random_bytes
import matplotlib.pyplot as plt

def plot_histogram(channel, title):
    plt.hist(channel.flatten(), bins=256, range=(0, 256), color='gray')
    plt.title(title)
    plt.xlabel('Pixel Value')
    plt.ylabel('Frequency')
    plt.grid(True)
    plt.show()

def calculate_entropy(channel):
    hist, _ = np.histogram(channel.flatten(), bins=256, range=(0, 256))
    prob = hist / np.sum(hist)
    return -np.sum(prob * np.log2(prob + 1e-12))

def calculate_mse(original, decrypted):
    return np.mean((original.astype(np.float64) - decrypted.astype(np.float64)) ** 2)

def calculate_npcr(original, encrypted):
    return np.sum(original != encrypted) / original.size * 100

def image_to_array(path):
    img = Image.open(path).convert('RGB')
```

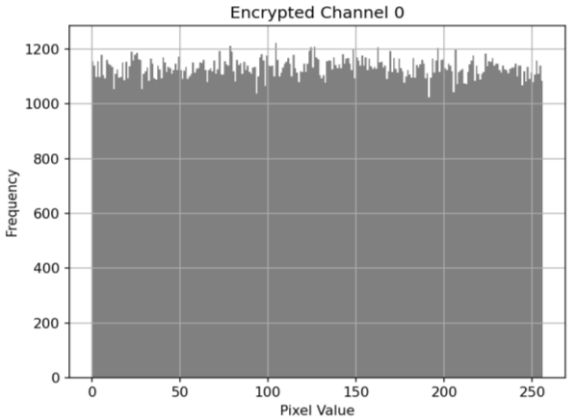
```
    arr = np.array(img)
    return arr, img.size, arr.shape

def array_to_image(array, shape, path):
    img = Image.fromarray(array.astype(np.uint8).reshape(shape))
    img.save(path)

def aes_encrypt_channel(channel, key, nonce):
    cipher = AES.new(key, AES.MODE_CTR, nonce=nonce)
    padded = pad(channel.tobytes(), AES.block_size)
    encrypted = cipher.encrypt(padded)
    return encrypted, cipher.nonce

def aes_decrypt_channel(encrypted, key, nonce, shape):
    cipher = AES.new(key, AES.MODE_CTR, nonce=nonce)
    decrypted = cipher.decrypt(encrypted)
    unpadded = unpad(decrypted, AES.block_size)
    return np.frombuffer(unpadded, dtype=np.uint8).reshape(shape)

def main(image_path):
    print(f"[+] Loading RGB image: {image_path}")
    img_array, size, shape = image_to_array(image_path)

    channels = [img_array[:, :, i] for i in range(3)]
    keys = [get_random_bytes(32) for _ in range(3)]
    nonces = [get_random_bytes(8) for _ in range(3)]

    encrypted_channels = []
    decrypted_channels = []
```

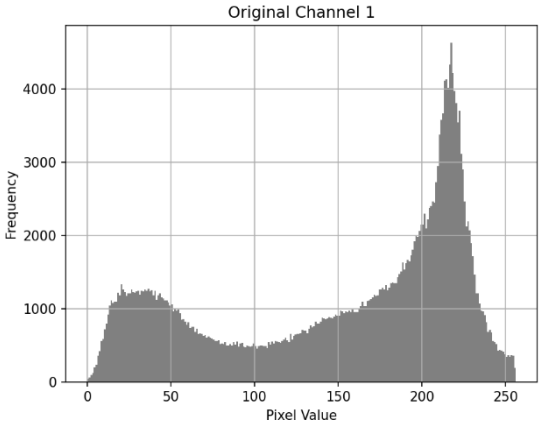
```
    for i in range(3):
        print(f"[+] Encrypting channel {i}")
        encrypted_bytes, nonce_used = aes_encrypt_channel(channels[i], keys[i], nonces[i])
        decrypted = aes_decrypt_channel(encrypted_bytes, keys[i], nonce_used,
channels[i].shape)

        encrypted = np.frombuffer(encrypted_bytes,
dtype=np.uint8)[:channels[i].size].reshape(channels[i].shape)
        encrypted_channels.append(encrypted)
        decrypted_channels.append(decrypted)

        entropy = calculate_entropy(encrypted)
        mse = calculate_mse(channels[i],encrypted)
        npcr = calculate_npcr(channels[i], encrypted)

        print(f"   Entropy: {entropy:.5f}")
        print(f"   MSE (after mod): {mse:.2f}")
        print(f"   NPCR: {npcr:.2f}%")

        plot_histogram(channels[i], f'Original Channel {i}')
        plot_histogram(encrypted, f'Encrypted Channel {i}')

    encrypted_img = np.stack(encrypted_channels, axis=2)
    decrypted_img = np.stack(decrypted_channels, axis=2)

    array_to_image(encrypted_img, shape, 'encrypted_rgb.png')
    array_to_image(decrypted_img, shape, 'decrypted_rgb.png')
    print("[+] Encrypted RGB image saved as 'encrypted_rgb.png'")
    print("[+] Decrypted RGB image saved as 'decrypted_rgb.png'")
```

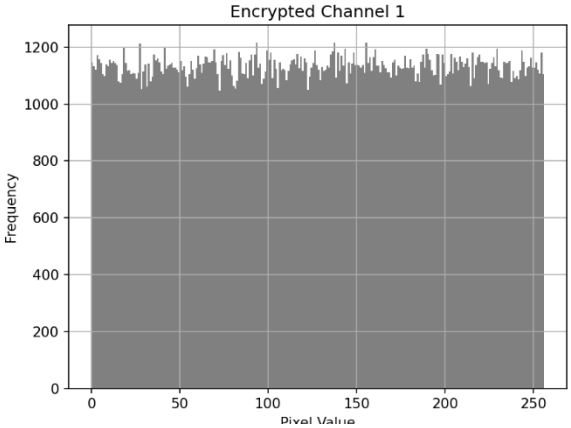
```
if __name__ == '__main__':
    main('first.png')
```

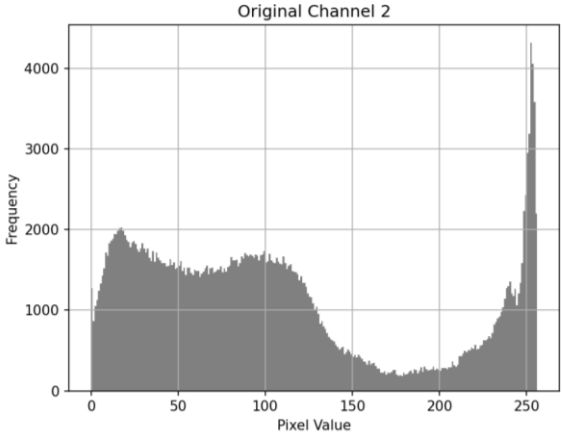
Output:

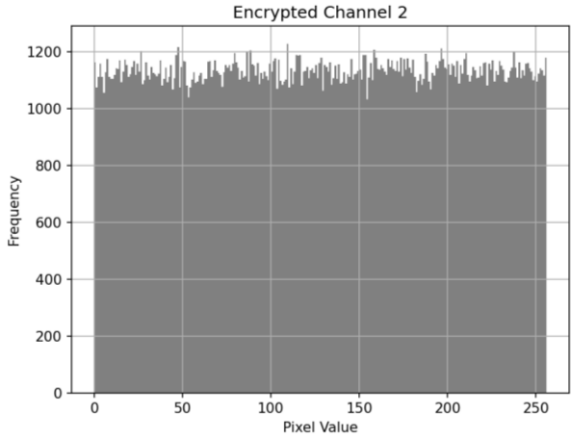








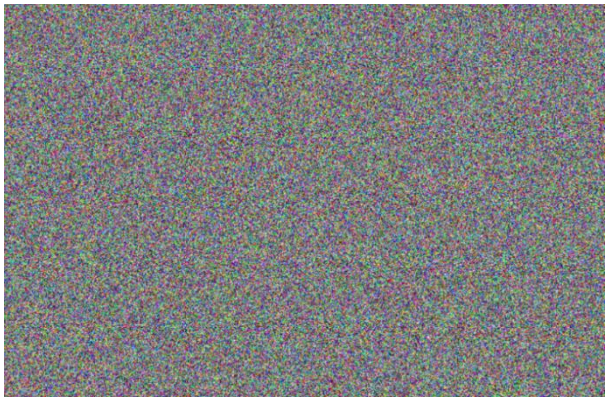




```
PS C:\Users\acer\Desktop\python> python proj1.py
[+] Loading RGB image: second.png
[+] Encrypting channel 0
    Entropy: 7.99942
    MSE : 9985.81
    NPCR: 99.62%
[+] Encrypting channel 1
    Entropy: 7.99938
    MSE : 11406.53
    NPCR: 99.64%
[+] Encrypting channel 2
    Entropy: 7.99943
    MSE : 12080.29
    NPCR: 99.60%
[+] Encrypted RGB image saved as 'encrypted_rgb.png'
[+] Decrypted RGB image saved as 'decrypted_rgb.png'
```

Input image:

Encrypted image:



Decrypted image:

Expected results:

Encrypted Image: Appears as noise with no identifiable visual patterns.

Decrypted Image: Nearly identical to the original, with minor simulated modification.

Entropy: Close to 8 for encrypted images, indicating strong randomness.

MSE: Slight increase due to simulated modification.

NPCR: Greater than 99%, showing high sensitivity and secure encryption.

Inference:

The implementation of AES-256 in CTR mode has proven to be highly effective for image encryption. The encrypted images exhibited entropy values close to the maximum (≈8), indicating a high level of randomness and unpredictability. Histogram analysis showed a uniform distribution of pixel intensities, confirming that the encrypted images do not retain any visible patterns from the original, which is essential for resisting visual and statistical attacks.

The NPCR values consistently above 99% demonstrated strong sensitivity to small changes in the original image—an important property for secure encryption. Additionally, the MSE values remained low (even after a small simulated modification), indicating that the decryption process was accurate and preserved the quality of the original image.

Overall, AES-256 in CTR mode ensures strong data confidentiality, making it suitable for secure image transmission in applications where privacy and data integrity are critical.

Reference:

T. Chandel, N. Gangodkar, A. Gautam, Y. Lohumi, M. R. Ghazi and A. Goyal, "Image Cryptography using AES-256 Algorithm," 2024 International Conference on Distributed Systems, Computer Networks and Cybersecurity (ICDSCNC), Bengaluru, India, 2024, pp. 1-6, doi: 10.1109/ICDSCNC62492.2024.10939131.